Assessment of Water Resources Availability in Krishna Upper Basin of India with SWAT Model using CORDEX Climate Projections

Bhaskara Rao[1], K.V. Rao[2], G.V. Srinivasa Reddy[3], M. Udaya Bhanu[4]

[1]ANGRAU- Agricultural Research Station, Anantapuramu - 515 701, Andhra Pradesh

[2]ICAR-Central Research Institute for Dryland Agriculture, Hyderabad - 500 059, Telangana

[3]College of Agricultural Engineering, UAS, Raichur - 584 104, Karnataka

[4]National Institute of Plant Health Management, Hyderabad - 500 030, Telangana

E-mail: bhaskararaoi@gmail.com

ABSTRACT: The impact of climate change on water balance components needs to be quantified from regional to local scale to manage water resources and suggest adaptation strategies to cope with future challenges. The present study was carried out to understand the climate change impact on water resources available at Krishna upper basin of India. CNRM RCM model was used to simulate the projections of rainfall and temperature with two RCP scenarios RCP 4.5 and RCP 8.5 and three time periods 2020-40 (near century), 2041-70 (mid-century) and 2071-99 (end century). It was observed that the change in mean annual rainfall projected by CNRM model under RCP 4.5 during 2020-40, 2041-70 and 2071-99 were 11.7, 26.7 and 32.2%, respectively and 18.6, 34.5 and 41.5%, respectively under RCP 8.5 scenarios. The CNRM model under RCP 4.5 scenario projected an increase in annual mean temperature for the period 2020-2040, 2041-2070 and 2071-2099 by 0.65, 1.06 and 1.49°C, respectively and 0.72, 1.55 and 2.41°C, under RCP 8.5 respectively. The SWAT model simulated that the projected annual surface runoff is likely to increase by 18, 70.2 and 102.7% under RCP 4.5 and 159.3, 106.9 and 66.2% under RCP 8.5 scenario during the future time slices of 2020-40, 2041-70 and 2071-99, respectively with reference to baseline period (1961-90) of surface runoff 195 mm under CNRM model. The annual surface runoff during the future periods is found to increase by 19.0, 50.0 and 27.0% under RCP 4.5 and 43.1, 100.5, 55.6% under RCP 8.5 scenario during 2020-40, 2041-70 and 2071-99 periods. It is concluded that the rainfall during the monsoon months except for September may increase in future periods and water availability in all the sub basins is expected to increase in future periods.

Keywords: Climate change, CORDEX, RCP, SWAT model, water availability

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Introduction

Water plays an important role in making the agriculture more sustainable. Agriculture sector consumes major portion of water resources of India (85-90%) as well as Global (70%) (Jain, 2021). The periodical assessment of water resources availability and demand with reference to time and space is required. Hydrological models can be used to assess the water resources in present and future time periods. India receives an annual precipitation about 4000 km^3 which is highly variable with space and time. The annual flow to river systems in India is about 1953 km^3 and replenishable groundwater resources assessed as 432 km^3. Out of these surface and groundwater resources, the utilizable surface and groundwater resources are 690 km^3 and 396 km^3 respectively (Kumar *et al.*, 2005). The total utilizable water resources of India are about 1123 km^2 (Billion Cubic Meter) and the per capita water availability is about 1651 m^3 (CWC, 2021). Agriculture sector utilizes major portion of water resources in India. It was estimated that the volume of water used for irrigation by last century was 428 km^3 (Kumar *et al.*, 2005). Among the major basins in India, utilizable water resources per year in Ganga basin are 250 km^3 followed by Godavari basin with 76 km^3. The utilizable water resources in Krishna basin are 58 km^3 against the potential resources of 89 km^3 (CWC, 2021). The water scarcity situation for various uses may still become worse if anticipated impacts of climate change on hydrology and water resources are considered.

According to the Intergovernmental Panel on Climate Change (Pachauri and Meyer, 2014), the global mean temperature rise may increase up to 4°C by the year 2100 and will severely affect the water resources availability and the water needs across the globe. Tangang *et al.* (2020) used Coordinated Regional Downscaling Experiment-South Asia (CORDEX-SA) data sets and projected extremes of precipitation for the end of 21st century (2081–2100) over Southeast Asia with magnitude of increase by 20% under Representative Concentration Pathway (RCP) 4.5 scenario and 30% under RCP8.5 scenario. Many studies reported that India would face frequent extreme weather conditions like wetter and dry conditions. Kumar *et al*. (2010) projected that air temperatures in India during the late 21st century will very likely record the extreme temperatures. Chanapathi *et al.* (2018) also used CORDEX-SA climate change data sets and reported that the temperature rise in Krishna basin is about 1.44°C (RCP 4.5) and 2.6°C (RCP 8.5) by end of the century. Nikam *et al.* (2018) also projected the rise in mean temperature by 0.021°C/year (RCP 4.5) and 0.045°C/year (RCP 8.5) and also reported increase in rainfall by 2 mm/year (RCP 4.5) and 3.85 mm/year

(RCP 8.5). Hydrological models can be used for quantification of water resources in the present prevailing situations and also be used for impact of climate change on hydrological components conditions. In the recent past, many researchers used lumped, distributed and semi distributed models based on conceptual and physical theory in nature to study the water resource availability at large river basin to micro watershed level. SWAT model is the most commonly used semi distributed, continues time scale hydrological model (Arnold *et al.*, 1998). Many researchers used SWAT model across the globe for evaluating the water availability and water balance components such as runoff, evapotranspiration, percolation, groundwater and lateral flow at basin level to micro watershed level (Jimeno *et al.*, 2018, Tabares *et al.*, 2019). The Krishna river basin is vulnerable to climate change, owing to the erratic distribution of rainfall along with warmer climatic conditions (Chanapathi *et al.*, 2018). Many researchers have carried out studies on climate change impact on water resources; few of them used CORDEX-SA datasets but did not sufficiently address the water availability in upper Krishna basin. With this background, this research work was carried out to assess the impact of climate change on water resources availability in future.

Materials and Methods

The present study was conducted on Krishna upper sub-basin which comprises of Upper Krishna (K1), Middle Krishna (K2), Ghataprabha (K3) and Malaprabha (K4) sub-basins. The study area is situated between 15°3′to 18°6′ North latitudes and 73°39′to77°23′ East longitudes, extended in the states of Maharashtra and Karnataka. The total drainage area of the study area is about 54,228 km^2 and has main outlet at Huvenhedigi in Raichur district of Karnataka. The elevation of study area ranges from 343 m to 1456 m with elevation difference of 1113 m. Geologically, the study area comprises of Deccan basaltic lava flows for upper cretaceous to lower Eocene age. Vertisols and laterites dominate soils in the basins (Deshpande, 1998). The high-altitude regions of the Western Ghats towards the western part of the study area experience more rainfall and the eastern part of the study area experiences semi-arid climate. The laterites are mostly found in high rainfall regions of Western Maharashtra and in the summits of the Deccan Traps. The location map of study area is given in Fig 1.

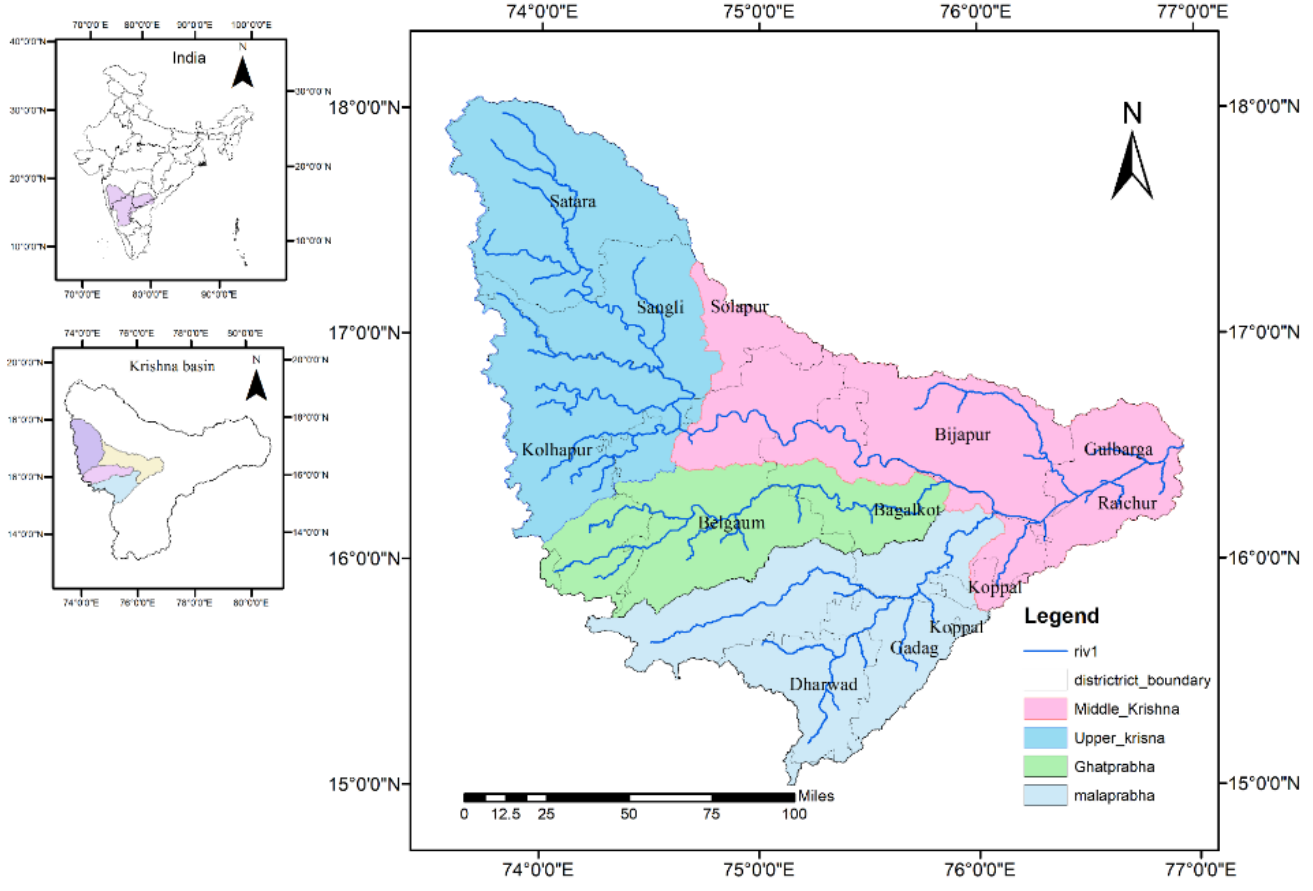


Fig. 1: Location map of the study area

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Analysis of rainfall and temperatures of CNRM for future periods

The dataset generated by CCCR-IITM (Centre for Climate Change Research-Indian Institute of Tropical Meteorology) using RegCM4 RCM were downloaded from http://cccr.tropmet.res.in/home/esgf_node.jsp CCCR-IITM ESGF data node which is high resolution (50 km) and projection is up to 2099. A total of 25 grid points with resolution of 0.44° x 0.44° (50 km) covering the study area were used for simulation of expected future climate change scenarios. The downloaded data provided a historical data from 1951 to 2005 and future projections from the year 2006 to 2099. Bias correction was implemented to adjust the data towards mean, standard deviation and variance. It was carried out for rainfall through multiplicative linear scale method and for minimum and maximum temperature through additive linear scale method (Shrestha *et al*., 2017; Dubey *et al.,* 2020). CMHyd tool was used for bias correction in climate data. Comparison of bias corrected historical average monthly rainfall data simulated from RCM models and observed data of IMD from the year 1951-2005.

Analysis of monthly and annual rainfall was carried out for the CNRM model at monthly and yearly time steps (Rao *et al*., 2021) The mean monthly and annual rainfall of future periods under CNRM climate model with two RCPs (RCP 4.5 and RCP 8.5) were compared with baseline period (1961-1990). The future time period (2020-2099) was divided into three time slices, namely, near century (2020-2040), midcentury (2041-2070) and end century (2071-2099). The mean monthly and annual rainfall of these times slices were also compared with the baseline period (1961-1990). The monthly maximum and minimum temperatures of CNRM model were also compared with the baseline data and the above same procedure was followed to analyze the temperature data as well.

SWAT model setup

ArcSWAT version 2012 was used for the study which is compatible and can add as a plugin with ArcGIS 10.3 version. The model set in the study focused on assessment of different water balance components and defines water availability with respect to space and time. The model was calibrated, validated with SWAT-CUP and was then used to assess climate change impact on water resources for future scenarios by selecting RCM models and by keeping the other variables constant. The detailed model inputs and setup was discussed in the paper published by Rao *et al.* (2023b). The calibrated and validated SWAT model was used to estimate the monthly water balance components at study area level and as well as major sub basins level. The historical water balance (1961-2019) using IMD gridded data was divided into two time slices. The present water balance and mean monthly water balance components were drawn from the average monthly water balance from the year 1991-2019. The monthly water balance components from the year 1961-1990 used as baseline period and compared with future time slicing periods such as 2020-2040, 2041-2070, 2071-2099 under climate change scenarios.

Water balance with SWAT model under RCMs with different scenarios

The calibrated and validated SWAT model was run with CNRM RCM model and two RCPs, RCP 4.5 and RCP 8.5 for the three future time periods (2020-40, 2041-2070, 2071-2099). The water balance components like surface runoff (SURQ), evapotranspiration (ET) and water yield (WYLD) were compared with baseline period. The water balance analysis was carried out at study area level, major sub-basin level and sub basin level. The spatial variability of key water balance components was also analyzed. The stream flow prediction was carried out at main outlet and the inflow prediction for the major irrigation projects (Almatti, Narayanpur, Ghataprabha (Hidkal) and Malaprabha (Renukasagar) dams) were also carried out for the future periods and compared with the baseline period.

Results and Discussion

Future rainfall projections

The projected mean annual rainfall of climate model CNRM under RCP 4.5 was found to be 922, against the observed baseline period (IMD (1961-1990) with rainfall of 754 mm. The projected mean annual rainfall under RCP 8.5 was on higher side with 968 mm, for CNRM (Fig. 2). The per cent change in monthly and annual rainfall projected by CNRM with RCM 4.5 and RCM 8.5 scenarios for different time periods are presented in Fig 3 (a) & (b). The changes in mean annual rainfall under RCP 4.5 during 2020-40, 2041-70 and 2071-99 were 11.7, 26.7 and 32.2% respectively and under RCP 8.5 scenarios, it was 18.6, 34.5 and 41.5% respectively. The monthly rainfall under RCP 4.5 and RCP 8.5 scenarios for all future periods (2020-40, 2041-70, 2071-99) during the monsoon months of June, July and August and in other season, April and October are expected to increase by 6 to 57%. The change in average monthly rainfall under RCP 4.5 of monsoon months June, July, August and September during the time period 2020-40 was 29, 6.4, 38.6 and -14.3% respectively and under RCP 8.5 scenario, it was 24.3, 28.1, 45.3 and -8.8%, respectively. The change in average monthly rainfall during the midcentury (2041-70) is expected to be more than the early century (2020-2041). During the period 2040-70, under RCP 4.5 scenario the change in average monsoon monthly rainfall in the months of June, July, August, and September was 46.8, 22.6, 42.4 and -0.1% respectively and under RCP 8.5 scenario, it was 24.3, 28.2, 45.5 and -8.7%, respectively. The projected change under RCP 4.5 scenario in

monthly average rainfall of June, July, August, and September during the period 2071-99 was found to be 37.4, 52, 24 and 11% respectively and under RCP 8.5 scenario, it was 17, 52.5, 92.7 and 6%, respectively. The average monthly rainfall under both the scenarios for February and March is expected to decrease in all the time slicing periods.

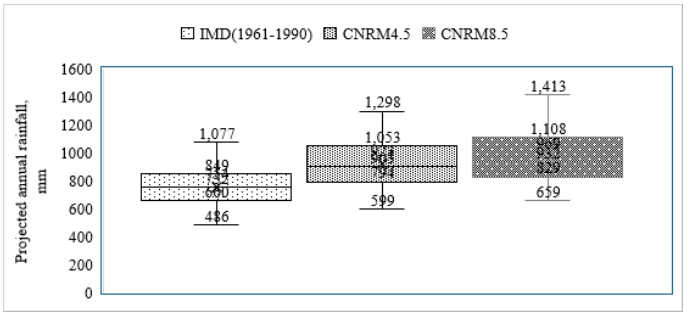
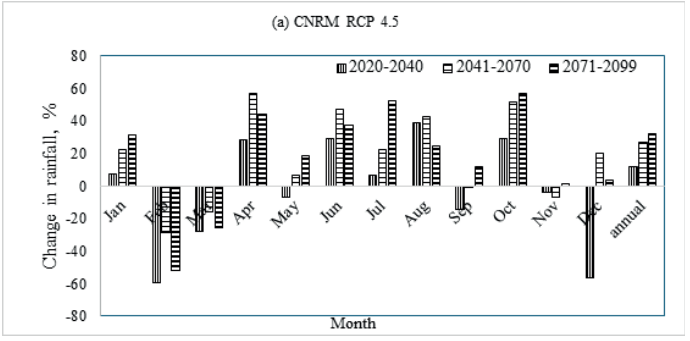


Fig. 2: Comparison of projected annual rainfall (2020-2099) under different climate models with baseline (IMD) data



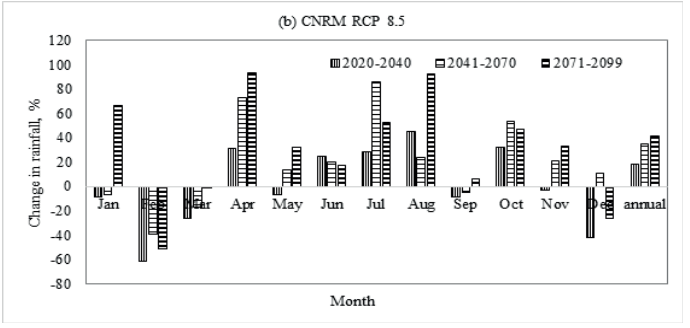
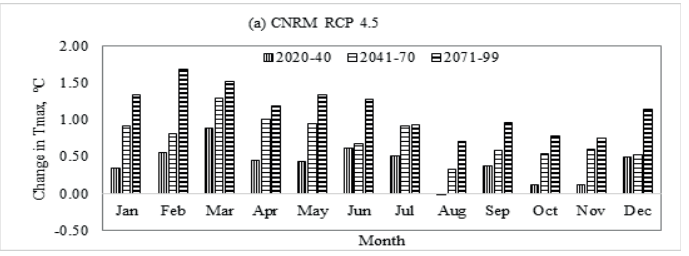


Fig. 3: Change in monthly rainfall under (a) RCP 4.5 and (b) RCP 8.5 for different time slicing periods

Effect of climate change on temperatures

The annual mean, maximum and minimum temperatures for the baseline period over the study were 26.3, 31.6 and 20.9°C, respectively. All the three models projected an increase in annual mean, maximum and minimum temperatures for all the three periods and under both RCP scenarios. The CNRM model under RCP 4.5 scenario projected an increase in annual mean temperature for the period 2020-2040, 2041-2070 and 2071-2099 by 0.65, 1.06 and 1.49°C, respectively and under RCP 8.5 it was 0.72, 1.55 and 2.41°C, respectively (Fig. 4). Chanapathi *et al.* (2018) reported similar results with CNRM model for entire Krishna basin.



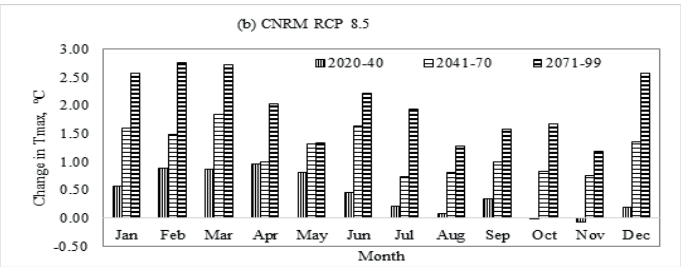


Fig. 4: Change in monthly maximum temperature under (a) RCP 4.5 and (b) RCP 8.5

Effect of climate change on water balance components

The average annual water balance components of study area are presented in the Table 1. The average annual rainfall, surface runoff, groundwater flow, evapo-transpiration and total water yield during the baseline period (1961-1990) were estimated as 754, 195, 102, 412 and 313 mm, respectively. CNRM model projected the annual surface runoff during the periods 2020-40, 2041-70 and 2071-99 as 230, 332 and 395 mm under RCP 4.5 scenario and 506, 404 and 324 mm under RCP 8.5 scenario. The projected annual surface runoff is likely to increase with 18, 70.2 and 102.7% under RCP 4.5 and 159.3, 106.9 and 66.2% under RCP 8.5 scenario during the future time slices 2020-40, 2041-70 and 2071-99 respectively with reference to baseline period. The increase in annual surface runoff was due to increase in annual rainfall by 40.1, 34 and 40.6% under RCP 4.5 scenario and 48.4, 43.5 and 38.9% under RCP 8.5 scenario during the periods 2020-40, 2041-70 and 2071-99 respectively with reference to rainfall during baseline period. Similar studies were conducted by several researchers (Chanapathi *et al.*, 2018; Nikam *et al.*, 2018; Chanapathi and Thatikonda, 2020) on Krishna basin and reported similar results. However, they reported that the evapotranspiration in future may decline with some climate models irrespective of change in rainfall, minimum and maximum temperatures. They reported that increasing trend in rainfall intensity for future scenarios may cause decline in evapotranspiration. The present study revealed that the evapotranspiration in future is likely to increase against the baseline period by 6.1 to 12.9% and 5.8 to 16% under RCP 4.5 and RCP 8.5 respectively. Swelam *et al.* (2010) reported that rise in temperature for future periods could lead to increase in evapotranspiration that could be compensated by reduction in transpiration by leaf stomata because of increase of CO_2 concentration with high humidity.

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Table 1: Annual water balance components of different RCMs model in different time periods

Parameter	Time period	Average annual water balance component (mm)			% change	
		Base period	CNRM 4.5	CNRM 8.5	CNRM 4.5	CNRM 8.5
Rainfall	2020-41	754	1057	1120	40.1	48.4
	2041-70		1014	1083	34.4	43.5
	2071-99		1061	1048	40.6	38.9
SurQ	2020-41	195	230	506	18.0	159.3
	2041-70		332	404	70.2	106.9
	2071-99		395	324	102.7	66.2
GWQ	2020-40	102	127	155	24.8	52.3
	2041-70		158	179	54.8	75.8
	2071-99		178	201	75.2	97.9
ET	2020-41	412	438	436	6.1	5.8
	2041-70		457	461	10.7	11.8
	2071-99		466	478	12.9	16.0
WYLD	2020-41	313	366	661	16.8	110.8
	2041-70		511	599	63.2	91.1
	2071-99		592	546	89.0	74.3

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Water yield projected by CNRM is likely to increase by 16.8%, 63.2% and 89.0% under RCM 4.5 scenario and 110.8, 91.1 and 74.3% under RCM 8.5 scenario during the time slice 2020-40, 2041-70 and 2071-99, respectively. Annual surface runoff during the baseline period ranged from 86.84 to 362.3 mm with an average of 195 mm. Annual surface runoff projected by CNRM model under RCP 4.5 and RCP 8.5 scenarios ranged from 119.5 to 552 mm and 133.5 to 709.7 mm with an average of 328.5 and 412.9 mm respectively. The water yield for study area during baseline period ranged from 128.3 to 530 mm with an average of 313.3 mm. The projected annual water yield by CNRM is found to be 186.8 to 836.5 mm with an average of 502.4 mm under the RPC 4.5 scenario and 224.4 to 968 mm with an average of 606 mm under RCP 8.5 scenario.

Effect of climate change on sub basin-wise surface water availability

The sub basin wise spatial distribution of average annual surface runoff (SURQ) for the baseline period (1960-90) and simulated with different climate models with both RCPs (RCP 4.5 and RCP 8.5) varied significantly. The sub-basins were further summarized based on the major sub basins of study area *i.e* Upper Krishna, Middle Krishna, Ghataprabha, Malaprabha sub basins. Surface runoff of sub-basins during the baseline period was observed between 52 to 994 mm. The Western parts of Upper Krishna and Ghataprabha produced more surface runoff (313 to 994 mm). These areas are influenced by Western Ghats that receive more rainfall. The average annual surface runoff of the sub-basins which were located in the Middle Krishna and Malaprabha and downstream part of Ghataprbha major sub basins ranged from 52 to 181 mm. The projected average annual runoff by CNRM model with RCP 4.5 during the periods 2020-40, 2041-70 and 2071-99 ranged from 47 to 1274, 74 to 1593 and 126 to 1848 mm, respectively. For the period 2020-40, it was observed that the projected surface runoff is likely to decrease due to projected decrease in rainfall. CNRM model under RCP 8.5 scenario projected the average annual stream flow for sub basins ranging from 70 to 1730, 77 to 1794 and 57 to 1770 mm during the time periods 2020-40, 2041-70 and 2071-99, respectively.

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Effect of climate change on monthly changes in surface runoff

The projected changes in monthly surface runoff with different climate change scenarios at study area level are presented in Table 2. Considerable change in projected monthly surface runoff was observed during the monsoon months (June, July and August) with all the climate model scenarios for all the periods. Highest change in monthly surface runoff was observed during the month of August with CNRM model under RCP 8.5 scenario during the period 2020-40. It was also observed that the surface runoff is likely to decrease in the month of September. CRNM model projected that the monthly surface runoff during October is likely to increase with CNRM model ranging from 2.8 to 22.3 mm under RCP 4.5 scenario and 3.6 to 23 mm under RCP 8.5 scenario.

Table 2: Change in average monthly surface runoff (in mm) in different time periods under different RCP scenarios against reference period

Month	CNRM 4.5			CNRM 8.5		
	2020-40	2041-70	2071-98	2020-40	2041-70	2071-98
January	-0.4	-0.4	-0.3	-0.4	-0.3	-0.2
February	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5
March	-0.5	-0.6	-0.6	-0.5	-0.6	-0.5
April	0.9	1.8	1.0	0.1	2.3	3.4
May	-2.5	0.2	-0.1	-3.7	2.3	1.0
June	16.4	56.4	17.4	15.7	70.1	5.6
July	6.9	31.8	147.6	120.3	108.3	54.9
August	28.7	37.2	16.5	169.8	17.1	64.4
September	-18.4	-2.3	0.9	-8.9	-10.5	-5.8
October	2.8	21.3	22.3	6.3	23.0	5.7
November	-1.9	-3.1	-2.8	-2.9	-1.5	2.1
December	-1.6	-0.5	-1.1	-1.6	-0.6	-1.1

Conclusion

It was observed that all the models projected that the mean annual rainfall is expected to increase during the three slicing periods under RCP 8.5 scenario than that of RCP 4.5 scenario. The projected annual rainfall by CNRM model was found to be higher in all the scenarios and future slicing periods. The CNRM model under RCP 4.5 scenario projected an increase in annual mean temperature for the period 2020-2040, 2041-2070 and 2071-2099 by 0.65, 1.06 and 1.49°C respectively and 0.72, 1.55 and 2.41°C under RCP 8.5 respectively. The SWAT model simulated that the projected annual surface runoff is likely to increase with 18, 70.2 and 102.7% under RCP 4.5 and 159.3, 106.9 and 66.2% under RCP 8.5 scenario during the future time slices 2020-40, 2041-70 and 2071-99 respectively with reference to baseline period (1961-90) surface runoff of 195 mm under CNRM model. The annual surface runoff during the future periods was found to increase by 19.0, 50.0 and 27.0% under RCP 4.5 and 43.1, 100.5, 55.6 under RCP 8.5 scenario during 2020-40, 2041-70 and 2071-99 periods. Considerable increase in projected monthly surface runoff was observed during the monsoon months (June, July and August) with all the climate model scenarios for all the periods with highest increase in August month with CNRM model under RCP 8.5 scenario during the period 2020-40. Climate models projected that the rainfall during the monsoon months except for September is found to be increasing in future periods. The mean annual temperature is expected to increase in the study area. The monthly minimum and maximum temperatures are expected to increase in future periods with increasing time periods. It is found that water availability in all the sub basins is expected to increase in future periods during monsoon period.

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